UCab — Full-Stack MERN Cab Booking Platform

Solo Developer / Team Member: Shaik Sumiya Zainab | Project ID: N/A (Solo Track)

Phase 3: Project Design Phase — Proposed Solution Description

Project Name: Cab Booking (`UCab`)

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1. Comprehensive Solution Overview

UCab is an end-to-end full-stack web platform built on the MERN stack (`MongoDB`, `Express.js`, `React.js`, `Node.js`) that transforms how urban commuters book cab rides and how fleet operators dispatch vehicles. Unlike standard ride-booking clones that merely list generic vehicles, UCab introduces a multi-tier vehicle categorization system coupled with a transparent deterministic fare calculation engine, live GPS state tracking, and a robust dual-engine persistence layer.

2. Key Pillars of the Proposed Solution

Pillar 1: Multi-Category Fleet & Transparent Fare Calculation

Instead of a single generic cab rate, UCab categorizes its inventory into four clear vehicle classes:

- Mini (` ¹10/km`): Affordable hatchbacks designed for short solo commutes (`Base:  ¹40`, 4 seats).
- Sedan (` ¹12/km`): Executive sedans offering extra legroom (`Base:  ¹50`, 4 seats).
- SUV (` ¹18/km`): Multi-utility vehicles for family and group travel (`Base:  ¹80`, 6 seats).
- Luxury (` ¹35/km`): High-end business-class sedans (`Base:  ¹150`, 4 seats).

When a user selects their pickup and drop-off points, the frontend instantly computes the route distance (`km`) and displays the exact fare breakdown:

`Total Fare = Base Fare + (Distance * Price Per KM) - Promo Discount + Perks`

Pillar 2: In-Ride Customization & Social Responsibility Suite

UCab enables riders to personalize their journey prior to boarding:

- Ø<ß_ In-Ride Refreshments Package (`+ ¹50`): Automatically instructs the assigned driver to keep chilled mineral water and premium snacks ready inside the cabin.
- Ø=Ü– Road Safety & Driver Welfare Donation (`+ ¹20`): Empowers riders to round up their fare, contributing directly to an emergency healthcare and accident insurance fund for drivers.

Pillar 3: Real-Time Dispatch Tracking & Corporate Invoicing

- Live Status Tracker (`UserHome.jsx`): Once a booking is confirmed, the user experiences a visual 4-stage tracking indicator (`Booking Accepted` !' `Driver Assigned` !' `Cab in Transit` !' `Ride Completed`) accompanied by complete driver credentials (`Vikram Sharma - 4.9 & `, contact phone, and vehicle registration plate).
- QR Corporate Receipts (`ReceiptModal.jsx`): Commuters can visit their booking history at any time (`MyBookings.jsx`) and download formal thermal corporate invoices complete with an embedded QR Verification Code for seamless company tax expense reimbursement.

Pillar 4: Executive Admin Control & Fleet CRUD

For dispatch leads and fleet owners, UCab provides a centralized command dashboard (`AdminHome.jsx`) that displays live key performance indicators (Total Gross Revenue ` ¹`, Total Bookings, Active Users, and Operational Fleet Count). Administrators can effortlessly add new cab inventory (`AddCar.jsx`), edit pricing and number plates (`EditCar.jsx`), toggle vehicle operational status between `Available` and `Busy`, and update trip dispatch states (`AdminBookings.jsx`) across the platform.

Pillar 5: Zero-Config Dual-Engine Database Architecture

To guarantee that the application can be evaluated or demonstrated on any laptop or university lab workstation without external database dependencies, a self-contained persistence engine (`server/db/store.js`) was engineered. If the Express server detects that MongoDB is offline on startup, it automatically switches from Mongoose ORM queries to atomic file operations (`data.json`), ensuring zero crashes and 100% data preservation.